

Algebraic Geometry – Exercise Sheet 2

Week 2: Prime spectra, the structure sheaf and affine subschemes
Monday and Thursday afternoon sessions; Lectures 6–10

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 6 only

Exercise 1. The affine line as a spectrum

★ CORE

Assume that $\mathbb{k}$ is algebraically closed and let $X = \operatorname{Spec}(\mathbb{k}[t])$.

- (i) List the points of X .
- (ii) Determine which points are closed and compute the closure of each point.
- (iii) For a non-zero polynomial f , describe the closed subset $\mathbb{V}(f)$.

Exercise 2. The spectrum of the integers

★ CORE

Describe $\operatorname{Spec}(\mathbb{Z})$: its points, its closed points, and the closure of each point. For $n \neq 0$, determine $\mathbb{V}(n)$.

Exercise 3. Nilpotents are invisible topologically

★ CORE

Let

$$A = \mathbb{k}[\varepsilon]/(\varepsilon^2), \quad X = \operatorname{Spec}(A).$$

Determine the points and open subsets of X . Compare the underlying topological space with $\operatorname{Spec}(\mathbb{k})$, and explain why the two affine schemes are nevertheless different.

Exercise 4. A quotient map on spectra

★ CHOICE

Consider the quotient homomorphism

$$\mathbb{k}[t] \longrightarrow \mathbb{k}[t]/(t^2).$$

Describe the induced map on spectra and its image. Why does the image depend only on $\sqrt{(t^2)}$?

Exercise 5. Closed subsets in the node

★★ CHOICE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \operatorname{Spec}(A)$. Denote the classes of x, y by the same letters.

- (i) Show that $\mathbb{V}(x) \cup \mathbb{V}(y) = X$ and determine $\mathbb{V}(x) \cap \mathbb{V}(y)$.
- (ii) Compute the closures of the points (x) and (y) .
- (iii) Show that neither of these two points is closed.

Exercise 6. The affine line over a function field

★★ CHOICE

Let $K = \mathbb{C}(t)$ and let $X = \operatorname{Spec}(K[x])$.

- (i) Describe the points and the closed points of X .
- (ii) Which closed points are K -rational? Give a closed point which is not K -rational.
- (iii) Put $S = \mathbb{C}[t] \setminus \{0\}$. Use $K[x] = S^{-1}\mathbb{C}[t, x]$ to identify X with the prime ideals of $\mathbb{C}[t, x]$ lying over $(0) \in \operatorname{Spec}(\mathbb{C}[t])$.

Thursday afternoon — material available: Lectures 6–9

Exercise 7. Principal opens in the node

★★ CORE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \operatorname{Spec}(A)$.

- (i) Describe $D(x)$ and $D(y)$ as affine schemes.
- (ii) Show that $D(x) \cap D(y) = \emptyset$ and identify the point missing from $D(x) \cup D(y)$.
- (iii) Compute the residue fields at (x) , (y) and (x, y) .

Exercise 8. Sections and a stalk on the affine line★ **CORE**

Let $X = \operatorname{Spec}(\mathbb{k}[t])$ and $U = D(t(t-1))$.

- (i) Compute $\mathcal{O}_X(U)$.
- (ii) Compute the stalk $\mathcal{O}_{X,(t-a)}$ and its residue field.
- (iii) For which a does $(t-a)$ belong to U ?

Exercise 9. A subscheme of the affine line with two support points★★ **CORE**

Let

$$Z = \operatorname{Spec}(\mathbb{k}[t]/(t^3(t-1)^2)).$$

- (i) Determine the underlying topological space of Z .
- (ii) Decompose its coordinate ring as a product of two rings.
- (iii) Compare the two local pieces with the reduced points at 0 and 1.

Exercise 10. Scheme-theoretic unions and intersections★ **CORE**

In $\mathbb{A}_{\mathbb{k}}^2$, let

$$X = \mathbb{V}(x), \quad Y = \mathbb{V}(y), \quad Y' = \mathbb{V}(y^2).$$

Compute the ideals defining $X \cup Y$, $X \cap Y$, $X \cup Y'$ and $X \cap Y'$. Compare their underlying closed subsets.

Exercise 11. Products of rings and disjoint unions★★ **CHOICE**

Let B and C be non-zero rings.

- (i) Show that every prime ideal of $B \times C$ is of the form $\mathfrak{p} \times C$ or $B \times \mathfrak{q}$.
- (ii) Deduce that

$$\operatorname{Spec}(B \times C) \simeq \operatorname{Spec}(B) \amalg \operatorname{Spec}(C).$$

- (iii) Relate this to the idempotents $(1, 0)$ and $(0, 1)$.

After Friday's arithmetic lecture or homework**Exercise 12. Prime ideals in $\mathbb{Z}[\sqrt{3}]$** ★★ **CHOICE**

Let

$$A = \mathbb{Z}[s]/(s^2 - 3), \quad f : \operatorname{Spec}(A) \longrightarrow \operatorname{Spec}(\mathbb{Z}).$$

For a prime p , describe the points of $\operatorname{Spec}(A)$ mapping to (p) by factoring $s^2 - 3$ over $\mathbb{F}_p$. Treat separately $p = 2$, $p = 3$, and $p \neq 2, 3$.

Exercise 13. An arithmetic conic★★★ **OPTIONAL**

For

$$X = \operatorname{Spec}(\mathbb{Z}[x, y]/(x^2 + y^2 - 5)) \longrightarrow \operatorname{Spec}(\mathbb{Z}),$$

write down the equation obtained after reduction modulo a prime p . Determine for which p that equation is a square or has a singular point.